“The Essentials of Computer Organization and Architecture”

Unit 2 – Exercises and Solutions

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//2.1. Perform the following base conversions using subtraction or division-remainder:

a) $458_{10} = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\__{3}$

b) $677_{10} = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\__{5}$

c) $1518_{10} = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\__{7}$

d) $4401_{10} = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\__{9}$

//2.1. Perform the following base conversions using subtraction or division-remainder:

a) 458_{10} = ______________ $_3$
b) 677_{10} = ______________ $_5$
c) 1518_{10} = ______________ $_7$
d) 4401_{10} = ______________ $_9$

Answers:

a. 121222_3 b. 10202_5 c. 4266_7 d. 6030_9

//2.2. Perform the following base conversions using subtraction or division-remainder:

a) 588_{10} = ______________ $_{3}$

b) 2254_{10} = ______________ $_{5}$

c) 652_{10} = ______________ $_{7}$

d) 3104_{10} = ______________ $_{9}$

//2.2. Perform the following base conversions using subtraction or division-remainder:

a) $588_{10} = \underline{\hspace{4cm}}_3$

b) $2254_{10} = \underline{\hspace{4cm}}_5$

c) $652_{10} = \underline{\hspace{4cm}}_7$

d) $3104_{10} = \underline{\hspace{4cm}}_9$

Answers:

a. 210210_3 b. 33004_5 c. 1621_7 d. 4228_9

2.3. Convert the following decimal fractions to binary using 6 places to the right of the binary point. Provide the absolute and relative error of the conversion.

a) 26.78125
b) 194.03125
c) 298.796875
d) 16.124023435

2.3. Convert the following decimal fractions to binary using 6 places to the right of the binary point. Provide the absolute and relative error of the conversion.

a) 26.78125
b) 194.03125
c) 298.796875
d) 16.124023435

Answers:

a) 11010.110010 = 26.78125 (no error)
b) 11000010.000010 = 194.03125 (no error)
c) 100101010.110011 = 298.796875 (no error)
d) 10000.000111 = 16.**109375**
- absolute error = 16.124023435 - 16.**109375** = 0.014648435
- relative error = (0.014648435 / 16.124023435)*100 = 0,09…%

2.4. Convert the following decimal fractions to binary using 6 places to the right of the binary point. Provide the absolute and relative error of the conversion.

a) 25.84375
b) 57.55
c) 80.90625
d) 84.874023

2.4. Convert the following decimal fractions to binary using 6 places to the right of the binary point. Provide the absolute and relative error of the conversion.

a) 25.84375
b) 57.55
c) 80.90625
d) 84.874023

Answers:

a) $11001.11011\underline{0}$ = 25.84375 (no error)

b) 111001.100011 = 57.**546875**

- absolute error = 57.55 - 57.**546875** = 0.003125
- relative error = (0.003125 / 57.55)*100 = 0,0054…%

c) $1010000.11101\underline{0}$ = 80.90625 (no error)

d) 1010100.110111 = 84.**859375**

- absolute error = 84.874023 - 84.**859375** = 0.014648
- relative error = (0.014648 / 84.874023)*100 = 0,017…%

2.5. Represent the following decimal numbers in binary using 8-bit signed magnitude, one's complement and two's complement:

a) 77
b) -42
c) 119
d) -107

2.5. Represent the following decimal numbers in binary using 8-bit signed magnitude, one's complement and two's complement:

a) 77
b) -42
c) 119
d) -107

Answers:

a. Signed magnitude: 01001101
One's complement: 01001101
Two's complement: 01001101

b. Signed magnitude: 10101010
One's complement: 11010101
Two's complement: 11010110

c. Signed magnitude: 01110111
One's complement: 01110111
Two's complement: 01110111

d. Signed magnitude: 11101011
One's complement: 10010100
Two's complement: 10010101

2.6. Using a "word" of 3 bits, list all of the possible signed binary numbers and their decimal equivalents that are representable in:

a) Signed magnitude
b) One's complement
c) Two's complement

2.6. Using a "word" of 3 bits, list all of the possible signed binary numbers and their decimal equivalents that are representable in:

a) Signed magnitude
b) One's complement
c) Two's complement

Answers:

Combinations		Signed Magnitude	One's Complement (C1)	Two's Complement (C2)
Signal	Value			
0	00	+ 0	+ 0	0
0	01	+ 1	+ 1	+ 1
0	10	+ 2	+ 2	+ 2
0	11	+ 3	+ 3	+ 3
1	00	- 00 = - 0	- C1(00) = - 11 = - 3	- C2(00) = - [C1(00) + 1] = - (11 + 1) = - 100 = - 4
1	01	- 01 = - 1	- C1(01) = - 10 = - 2	- C2(01) = - [C1(01) + 1] = - (10 + 1) = - 11 = - 3
1	10	- 10 = - 2	- C1(10) = - 01 = - 1	- C2(10) = - [C1(10) + 1] = - (01 + 1) = - 10 = - 2
1	11	- 11 = - 3	- C1(11) = - 00 = - 0	- C2(11) = - [C1(11) + 1] = - (00 + 1) = - 01 = - 1

2.7. Using a "word" of 4 bits, list all of the possible signed binary numbers and their decimal equivalents that are representable in:

a) Signed magnitude
b) One's complement
c) Two's complement

2.7. Using a "word" of 4 bits, list all of the possible signed binary numbers and their decimal equivalents that are representable in:

a) Signed magnitude
b) One's complement
c) Two's complement

Answers:

a. 0111 to 1111, or +7 to -7
b. 0111 to 1000, or +7 to -7
c. 0111 to 1000, or +7 to -8

2.16. Using signed-magnitude representation, and assuming a 4 bit representation, complete the following operations:

a) $+0 + (-0) =$
b) $(-0) + 0 =$
c) $0 + 0 =$
d) $(-0) + (-0) =$

2.16. Using signed-magnitude representation, and assuming a 4 bit representation, complete the following operations:

a) + 0 + (- 0) =
b) (- 0) + 0 =
c) 0 + 0 =
d) (- 0) + (- 0) =

Answers:

```
  +0 + (-0) = 0000 + 1000 = 1000 (-0)
  (-0) + 0  = 1000 + 0000 = 1000 (-0)
     0 + 0  = 0000 + 0000 = 0000 (+0)
(-0) + (-0) = 1000 + 1000 = 0000 (+0)
```

2.18. If the floating-point number storage on a certain system has 1 sign bit, 3-bit exponent and 4-bit mantissa ...

... What is the largest positive and the smallest positive number that can be stored on this system if the storage is normalized? (Assume no bits are implied, there is no biasing, **exponents use two's complement**, exponents of all 0's and all 1's aren't allowed.)

2.18. If the floating-point number storage on a certain system has 1 sign bit, 3-bit exponent and 4-bit mantissa ...

... What is the largest positive and the smallest positive number that can be stored on this system if the storage is normalized? (Assume no bits are implied, there is no biasing, **exponents use two's complement**, exponents of all 0's and all 1's arent' allowed.)

Answer:

smallest positive:

- **exponent**: in C2, the smallest possible value with 3 bits is $-4_{10} = 100_2$, and this value is different from all 0's and all 1's (as required by the problem)
- **mantissa**: the smallest normalized value possible is $0.1\underline{000}$
- floating-point representation: | 0 | 100 | 1000 |
- normalized scientific binary representation: $\mathbf{0.1_2 \times 2^{-4}}$
- value: $\mathbf{0.1_2 \times 2^{-4}} = 0.01_2 \times 2^{-3} = \ldots = 0.00001_2 \times 2^0 = 2^{-5} = 1/32 = \mathbf{0.03125_{10}}$

2.18. If the floating-point number storage on a certain system has 1 sign bit, 3-bit exponent and 4-bit mantissa ...

... What is the largest positive and the smallest positive number that can be stored on this system if the storage is normalized?(Assume no bits are implied, there is no biasing, **exponents use two's complement**, exponents of all 0's and all 1's arent' allowed.)

Answer (cont.):

largest positive:

- **exponent**: in C2, the largest possible value with 3 bits is $+3_{10} = 011_2$, and this value is different from all 0's and all 1's (as required by the problem)
- **mantissa**: the largest normalized value possible is 0.1111
- floating-point representation: | 0 | 011 | 1111 |
- binary normalized scientific representation: $\mathbf{0.1111_2 \times 2^3}$
- value: $\mathbf{0.1111_2 \times 2^3} = 1.111_2 \times 2^2 = 11.11_2 \times 2^1 = 111.1_2 \times 2^0 = 7 + 0.5 = \mathbf{7.5_{10}}$

2.20. Assuming the 14-bit simple model for floating-point representation (1 sign bit, 5 bits for the exponent with a bias of 16, a normalized mantissa of 8 bit):

a) Show how the computer would represent **100.0** and **0.25**.
b) Show how the computer would add the two floating-point numbers of a) by changing one of the numbers so they are both expressed using the same power of 2.
c) Show how the computer would represent the sum of b) using the given floating-point representation. What decimal value for the sum is the computer actually storing?

2.20. Assuming the 14-bit simple model for floating-point representation (1 sign bit, 5 bits for the exponent with a bias of 16, a normalized mantissa of 8 bit):

a) Show how the computer would represent **100.0** and **0.25**.

Answer:

100.0

- $100_{10} = 64+32+4 = 1100100_2 \times 2^0 = 110010.0_2 \times 2^1 = \ldots = 0.1100100_2 \times 2^7$
- normalized **mantissa**, using 8 bits: $1100100\underline{0}_2$
- **exponent** with bias, using 5 bits: $16 + 7 = 16 + 4 + 2 + 1 = 10111_2$
- positive signal => signal bit is 0
- floating-point representation: | 0 | 10111 | 11001000 |

2.20. Assuming the 14-bit simple model for floating-point representation (1 sign bit, 5 bits for the exponent with a bias of 16, a normalized mantissa of 8 bit):

a) Show how the computer would represent 100.0 and 0.25.

Answer (cont.):

0.25:

- $0.25_{10} = 2^{-2} = 0.01_2 \times 2^0 = 0.1_2 \times 2^{-1}$
- normalized **mantissa**, using 8 bits: 10000000_2
- **exponent** with bias, using 5 bits: $16 + (-1) = 15 = 8+4+2+1 = 01111_2$
- positive signal => signal bit is 0
- floating-point representation: | 0 | 01111 | 10000000 |

2.20. Assuming the 14-bit simple model for floating-point representation (1 sign bit, 5 bits for the exponent with a bias of 16, a normalized mantissa of 8 bit):

b) + c)

Answer:

- for the mantissas to be summed, the exponents are required to match
- if they don't match, one of the numbers will have to be denormalized
- the number to be denormalized is the smallest number, once any loss of digits that may happen will have the smallest impact in the final value of the sum

- thus:
  - $0.25_{10} = 2^{-2} = 0.01_2 \times 2^0 = 0.001_2 \times 2^1 = \ldots = 0.000000001_2 \times 2^7$
  - denormalized **mantissa**, with 8 bits: $\underline{00000000_2}$
  - **exponent** with bias, using 5 bits: $16 + (-7) = 15 = 8 + 4 + 2 + 1 = 01111_2$
  - floating-point representation: | 0 | 10111 | 00000000 |

2.20. Assuming the 14-bit simple model for floating-point representation (1 sign bit, 5 bits for the exponent with a bias of 16, a normalized mantissa of 8 bit):

b) + c)

Answer (cont):

- the sum of both mantissas is now

	mantissa	
	11001000	(mantissa of 100)
+	00000000	(mantissa of 0.25)
	11001000	

- result in floating-point representation: | 0 | 10111 | **11001000** |
- **exponent** without bias: $10111_2 - 16_{10} = 23 - 16 = 7$
- result in binary normalized scientific notation:
  $0.\mathbf{11001000}_2 \times 2^7 = 1100100.0_2 \times 2^0 = 100.0_{10}$
- that is, the contribution of 0.25 was lost for the sum 100.0 + 0.25

2*.1. For $A=65.0_{10}$ and $B=50.0625_{10}$ Answer the following questions:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part{?} fractional part 0.{?}
Δ1={?} (value lost in conversion)

B = integer part{?} parte fractional part 0.{?}
Δ2={?} (value lost in conversion)

2*.1. For A=65.0_{10} and B=50.0625_{10} Answer the following questions:

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**1000001**} fractional part 0.{**0000**}
Δ1={**0**} (value lost in conversion)

B = integer part {**110010**} fractional part 0.{**0001**}
Δ2={**0**} (value lost in conversion)

2*.1. For A=65.0_{10} and B=50.0625_{10} Answer the following questions:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand). Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {?} | exponent {?} | significand {?}
Δ3={?} (value lost in representation)

B = sign {?} | exponent {?} | significand {?}
Δ4={?} (value lost in representation)

2*.1. For A=65.0_{10} and B=50.0625_{10} Answer the following questions:

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand). Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10111**} | significand {**10000010**}
Δ3={**0**} (value lost in representation)

B = sign {**0**} | exponent {**10110**} | significand {**11001000**}**01**
Δ4={**0.0625**} (value lost in representation)

2*.1. For $A=65.0_{10}$ and $B=50.0625_{10}$ Answer the following questions:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

$$
\begin{array}{ll}
 & \{?\}|\{?\}|\{?\} = A \\
\underline{+} & \underline{\{?\}|\{?\}|\{?\} = B} \\
 & \{?\}|\{?\}|\{?\} = A + B
\end{array}
$$

$\Delta 5=\{?\}$ (value lost through denormalization of the smallest part)

$\Delta 6=\{?\}$ (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{?}

Confirmation: $A + B = 2.c) + \Sigma\Delta$

2*.1. For $A=65.0_{10}$ and $B=50.0625_{10}$ Answer the following questions:

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

$$
\begin{array}{rl}
 & \{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{1000001\underline{0}}\} = A \\
+ & \underline{\{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{01100100}\} = B} \\
 & \{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{11100110}\} = A + B
\end{array}
$$

Δ5={**0**} (value lost through denormalization of the smallest part)

Δ6={**0**} (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{**115**}

Confirmation: 65 + 50.0625 = 115 + 0.0625

2*.2. Solve exercise 2*.1 again for A=101.0 and B=0.75

2*.2. Solve exercise 2*.1 again for A=101.0 and B=0.75

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**1100101**} fractional part 0.{**0000**}
Δ1={**0**} (value lost in conversion)

B = integer part {**0**} fractional part 0.{**1100**}
Δ2={**0**} (value lost in conversion)

2*.2. Solve exercise 2*.1 again for A=101.0 and B=0.75

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand).
Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10111**} | significand {**110010110**}
Δ3={**0**} (value lost in representation)

B = sign {**0**} | exponent {**10000**} | significand {**1100<u>0000</u>**}
Δ4={**0**} (value lost in representation)

2*.2. Solve exercise 2*.1 again for A=101.0 and B=0.75

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

$$
\begin{array}{rl}
 & \{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{11001010}\} = A \\
+ & \underline{\{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{00000001}\}\mathbf{1} = B} \\
 & \{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{11001011}\} = A + B
\end{array}
$$

$\Delta 5=\{\mathbf{0.25}\}$ (value lost through denormalization of the smallest part)

$\Delta 6=\{\mathbf{0}\}$ (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{**101.5**}

Confirmation: 101 + 0.75 = 101.5 + 0.25

2*.3. Solve exercise 2*.1 again for A=55.55 and B=7.777

2*.3. Solve exercise 2*.1 again for A=55.55 and B=7.777

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**110111**} fractional part 0.{**1000**} **...**
Δ1={**0.05**} (value lost in conversion)

B = integer part {**111**} fractional part 0.{**1100**} **...**
Δ2={**0.027**} (value lost in conversion)

2*.3. Solve exercise 2*.1 again for A=55.55 and B=7.777

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand).
Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10110**} | significand {**11011110**}
Δ3={**0**} (value lost in representation)

B = sign {**0**} | exponent {**10011**} | significand {**11111000**}
Δ4={**0**} (value lost in representation)

2*.3. Solve exercise 2*.1 again for A=55.55 and B=7.777

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

{**0**}|{**10110**}|{**11011110**} = A
\+ {**0**}|{**10110**}|{**00011111**} = B
{**0**}|{**10110**}|{**11111101**} = A + B

Δ5={**0**} (value lost through denormalization of the smallest part)
Δ6={**0**} (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{**63.25**} Confirmation: 55.55 + 7.777 = 63.25 + 0.05 + 0.027

2*.4. Solve exercise 2*.1 again for A=64.0 and B=75.75

2*.4. Solve exercise 2*.1 again for A=64.0 and B=75.75

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**1000000**} fractional part 0.{**0000**}
Δ1={**0**} (value lost in conversion)

B = integer part {**1001011**} fractional part 0.{**1100**}
Δ2={**0**} (value lost in conversion)

2*.4. Solve exercise 2*.1 again for A=64.0 and B=75.75

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand).
Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10111**} | significand {**10000000**}
Δ3={**0**} (value lost in representation)

B = sign {**0**} | exponent {**10111**} | significand {**10010111**}**1**
Δ4={**0.25**} (value lost in representation)

2*.4. Solve exercise 2*.1 again for A=64.0 and B=75.75

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum..

```
                     1
    {0}|{10111}|{10000000}  = A
+   {0}|{10111}|{10010111}  = B
--------------------------------
                111
             10111 | 00010111
                +1 | 100010111
--------------------------------
    {0}|{11000}|{10001011}  = A + B
```

Δ5={**0**} (value lost through denormalization of the smallest part)

Δ6={**0.5**} (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{**139**} Confirmation 64 + 75.75 = 139 + 0.25 + 0.5

2*.5. Solve exercise 2*.1 again for A=88.88 and B=9.999

2*.5. Solve exercise 2*.1 again for A=88.88 and B=9.999

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**1011000**} fractional part 0.{**1110**} **...**
Δ1={**0,005**} (value lost in conversion)

B = integer part {**1001**} fractional part 0.{**1111**} **...**
Δ2={**0,0615**} (value lost in conversion)

2*.5. Solve exercise 2*.1 again for A=88.88 and B=9.999

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand).
Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10111**} | significand {**10110001**}**11**
Δ3={**0,375**} (value lost in representation)

B = sign {**0**} | exponent {**10100**} | significand {**10011111**}
Δ4={**0**} (value lost in representation)

2*.5. Solve exercise 2*.1 again for A=88.88 and B=9.999

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

$$
\begin{array}{rl}
 & \phantom{\{0\}|\{10111\}|\{}\mathbf{11}\ \ \ \mathbf{11} \\
 & \{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{10110001}\} = A \\
+ & \underline{\{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{00010011}\}\mathbf{111} = B} \\
 & \{\mathbf{0}\}|\{\mathbf{10111}\}|\{\mathbf{11000100}\} = A + B
\end{array}
$$

Δ5={**0,4375**} (value lost through denormalization of the smallest part)

Δ6={**0**} (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{**98**} Confirmation: 88.88 + 9.99 = 98 + 0.05 + 0.0615 + 0.375 + 0.4375

2*.6. Solve exercise 2*.1 again for A=66.66 and B=99.99

2*.6. Solve exercise 2*.1 again for A=66.66 and B=99.99

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**1000010**} fractional part 0.{**1010**} **...**
Δ1={**0.035**} (value lost in conversion)

B = integer part {**1100011**} fractional part 0.{**1111**} **...**
Δ2={**0.0525**} (value lost in conversion)

2*.6. Solve exercise 2*.1 again for A=66.66 and B=99.99

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand).
Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10111**} | significand {**10000101**} **010**
Δ3={**0.125**} (value lost in representation)

B = sign {**0**} | exponent {**10111**} | significand {**11000111**} **111**
Δ4={**0.4375**} (value lost in representation)

2*.6. Solve exercise 2*.1 again for A=66.66 and B=99.99

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

```
                 1          111
   {0}|{10111}|{10000101}  =  A
+  {0}|{10111}|{11000111}  =  B
___________________________________
             111
          10111  |  01001100
              +1  |  101001100
___________________________________
   {0}|{11000}|{10100110}  =  A + B
```

Δ5={**0**} (value lost through denormalization of the smallest part)

Δ6={**0**} (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{**166**} Confirmation: 66.66 + 99.99 = 166 + 0.035 + 0.0525 + 0.125 + 0. 4375

2*.7. Solve exercise 2*.1 again for A=97.53 and B=35.79

2*.7. Solve exercise 2*.1 again for A=97.53 and B=35.79

Answer:

1. Convert A and B to base 2, considering a precision of 4 binary digits for the fractional part.

A = integer part {**1100001**} fractional part 0.{**1000**} **...**
$\Delta 1$={**0.03**} (value lost in conversion)

B = integer part {**100011**} fractional part 0.{**1100**} **...**
$\Delta 2$={**0.04**} (value lost in conversion)

2*.7. Solve exercise 2*.1 again for A=97.53 and B=35.79

Answer:

2. Consider the simplified 14-bit floating-point model (1 bit of sign, 5 bits of exponent with bias 16, and 8 bits of normalized significand). Using as a starting point the binary representations of A and B obtained in question 1:

2.a) Show how A and B would be represented in this model.

A = sign {**0**} | exponent {**10111**} | significand {**11000011**}
Δ3={**0**} (value lost in representation)

B = sign {**0**} | exponent {**10110**} | significand {**10001111**}
Δ4={**0**} (value lost in representation)

2*.7. Solve exercise 2*.1 again for A=97.53 and B=35.79

Answer:

2.b) Considering the sum A + B in this model, complete the representation of the parts and the result of sum.

```
                   1     1     111
   {0}|{10111}|{11000011}  = A
+  {0}|{10111}|{01000111}1  = B
--------------------------------------------
             111
          10111 | 00001010
              +1 | 100001010
--------------------------------------------
   {0}|{11000}|{10000101}  = A + B
```

Δ5={0.25} (value lost through denormalization of the smallest part)

Δ6={0} (value lost in the representation of the result)

2.c) Present the sum result value in base 10.

{133} Confirmation: 97.53 + 35.79 = 133 + 0.03 + 0.04 + 0.25

2.24. Conversions: ASCII and EBCDIC

a) Knowing that ASCII code for the character A is $1000001_2 = 65_{10}$, what is the ASCII code for the character J?

b) Knowing that EBCDIC code for the character A is $1100\ 0001_2$, what is the EBCDIC code for the character a ?

2.24. Conversion of ASCII and EBCDIC

a) Knowing that ASCII code for the character A is $1000001_2 = 65_{10}$, what is the ASCII code for the character J?

b) Knowing that EBCDIC code for the character A is $1100\ 0001_2$, what is the EBCDIC code for the character a ?

Answer:

a) $J = A + 9 = 74_{10} = 1001010_2$

b) it is enough to negate the 2nd most significant bit: a $=1\mathbf{0}00\ 0001_2$

2.25. Assume a 24-bit word on a computer. How would the computer represent the string “295”, with 7-bit ASCII and a) even parity, or b) odd parity ?

Help:

0	NUL	16	DLE	32		48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	TAB	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

2.25. Assume a 24-bit word on a computer. How would the computer represent the string “295”, with 7-bit ASCII and a) even parity, or b) odd parity ?

Answer:

a)

'2'=50	'9'=57	'5'=53
1 0 1 1 0 0 1 0	0 0 1 1 1 0 0 1	0 0 1 1 0 1 0 1

b)

'2'=50	'9'=57	'5'=53
0 0 1 1 0 0 1 0	1 0 1 1 1 0 0 1	1 0 1 1 0 1 0 1

2.26. Decode the following ASCII message, assuming 7-bit ASCII characters and no parity: 1001010 1001111 1001000 1001110 0100000 1000100 1001111 1000101

Help:

0	NUL	16	DLE	32		48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	TAB	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

2.26. Decode the following ASCII message, assuming 7-bit ASCII characters and no parity: 1001010 1001111 1001000 1001110 0100000 1000100 1001111 1000101

Answer:

100 1010 = J
100 1111 = O
100 1000 = H
100 1110 = N
010 0000 = space
100 0100 = D
100 1111 = O
100 0101 = E